

Diversity of Data Types for Data Mining (I)

- ❑ **Structured vs. unstructured data**
 - ❑ *Structured*: uniform, record- or table-like structures, defined by data dictionaries, with a fixed set of attributes, each with a fixed set of value ranges and semantic meaning
 - ❑ Ex. Data stored in *relational databases*, *data cubes*, *data matrices*, and many *data warehouses*
 - ❑ *Semi-structured*: allow a data object to contain a set value, a small set of heterogeneous typed values, or nested structures, or to allow the structure of objects or sub-objects to be defined flexibly and dynamically
 - ❑ Data having *certain structures* with clearly defined semantic meaning, such as *transactional data set*, *sequence data set* (e.g., time-series data, gene or protein data, or Weblog data)
 - ❑ *Graph or network data*: A more sophisticated type of semi-structured data set
 - ❑ *Unstructured data*: text data and multimedia (e.g., audio, image, video) data
- ❑ The real-world data can often be a mixture of structured, semi-structured data and unstructured data

Diversity of Data Types for Data Mining (II)

□ Data associated with different applications

- Different applications: different data sets and require different data analysis methods
 - Sequence data: *Biological sequences vs. shopping transaction sequences*
 - Time-series: ordered set of numerical values with equal time interval
 - Spatial, temporal and spatiotemporal data
 - Graph and network data: Social networks, computer communication networks, biological networks, and information networks may carry rather different semantics
- On the same data set, finding different kinds of patterns: require different mining methods
 - Ex. software programs: finding plagiarized modules vs. finding copy-and-paste bugs

□ Stored vs. streaming data

- Stored data: Finite, stored in various kinds of large data repositories
- Streaming data (e.g., video surveillance or remote sensing): Dynamic, constantly coming, infinite, real-time response—posing challenges on effective data mining